Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no............................

Name of the Course: MCA
Semester: II
Name of the Paper: Software Engineering
Paper Code: TMC 202
Time: 1.5-hour **Maximum Marks: 50**

Note:

(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Define software engineering. Explain the importance and characteristics of software. **(CO1)**

OR

b. Explain the principles of Requirements Engineering. Compare and analyze different requirements elicitation techniques used to collect user requirements. **(CO2)**

Q2. (10 Marks)

a. Discuss the Software Development Life Cycle (SDLC). Illustrate the stages and evaluate how each stage contributes to achieving software quality. **(CO1)**

OR

b. Illustrate the structure of a Software Requirement Specification (SRS). Differentiate between requirement specification, monitoring, and control, and justify their importance. **(CO2)**

Q3. (10 Marks)

a. Define software engineering problems and analyze the components and applications of software. **(CO1)**

OR

b. Explain the concept of Data Flow Diagrams (DFDs). Construct suitable DFDs and interpret their role in requirement analysis and good software design. **(CO2)**

Q4. (10 Marks)

a. Explain the concepts of Software Quality, Reliability, Usability, and Performance. Evaluate how these attributes influence the success of the software process. **(CO1)**

OR

b. Compare the structured (functional) design approach and object-oriented approach. Illustrate the fourth-generation techniques in software development. **(CO2)**

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Q5. (10 Marks)

a. Compare and evaluate Waterfall and Agile model and also discuss their advantages and limitations respectively. **(CO2)**

OR

b. Analyze the effect of cohesion and coupling on functional independence. **(CO2)**